

Advanced Internet Protocols, Services, and Applications

Eiji Oki • Roberto Rojas-Cessa
Mallikarjun Tatipamula • Christian Vogt

ADVANCED INTERNET PROTOCOLS, SERVICES, AND APPLICATIONS

ADVANCED INTERNET PROTOCOLS, SERVICES, AND APPLICATIONS

Eiji Oki

University of Electro-Communications, Japan

Roberto Rojas-Cessa

New Jersey Institute of Technology, USA

Mallikarjun Tatipamula

Ericsson Silicon Valley, USA

Christian Vogt

Ericsson Silicon Valley, USA

 **WILEY**

A JOHN WILEY & SONS, INC., PUBLICATION

Copyright © 2012 by John Wiley & Sons, Inc. All rights reserved

Published by John Wiley & Sons, Inc., Hoboken, New Jersey

Published simultaneously in Canada

No part of this publication may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning, or otherwise, except as permitted under Section 107 or 108 of the 1976 United States Copyright Act, without either the prior written permission of the Publisher, or authorization through payment of the appropriate per-copy fee to the Copyright Clearance Center, Inc., 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400, fax (978) 750-4470, or on the web at www.copyright.com. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008, or online at <http://www.wiley.com/go/permission>.

Limit of Liability/Disclaimer of Warranty: While the publisher and author have used their best efforts in preparing this book, they make no representations or warranties with respect to the accuracy or completeness of the contents of this book and specifically disclaim any implied warranties of merchantability or fitness for a particular purpose. No warranty may be created or extended by sales representatives or written sales materials. The advice and strategies contained herein may not be suitable for your situation. You should consult with a professional where appropriate. Neither the publisher nor author shall be liable for any loss of profit or any other commercial damages, including but not limited to special, incidental, consequential, or other damages.

For general information on our other products and services or for technical support, please contact our Customer Care Department within the United States at (800) 762-2974, outside the United States at (317) 572-3993 or fax (317) 572-4002.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic formats. For more information about Wiley products, visit our web site at www.wiley.com.

Library of Congress Cataloging-in-Publication Data:

Oki, Eiji, 1969–

Advanced Internet protocols, services, and applications / Eiji Oki, Roberto Rojas-Cessa, Mallikarjun Tatipamula, and Christian Vogt.

pages cm

Includes bibliographical references and index.

ISBN 978-0-470-49903-0

1. Computer network protocols. I. Rojas-Cessa, Roberto. II. Tatipamula, Mallikarjun.

III. Vogt, Christian (Marketing executive) IV. Title.

TK5105.55.O54 2012

004.6–dc23

2011048524

Printed in the United States of America

ISBN: 978-0-470-49903-0

10 9 8 7 6 5 4 3 2 1

CONTENTS

Preface	xi
Acknowledgments	xv
About the Authors	xvii
1 Transmission Control Protocol/Internet Protocol Overview	1
1.1 Fundamental Architecture / 1	
1.2 Internet Protocol Basics / 4	
1.2.1 Packet Header / 5	
1.2.2 Internet Protocol Address / 7	
1.2.3 Internet Protocol Classification / 7	
1.2.4 Subnet and its Masking / 9	
1.2.5 Subnet Calculation / 11	
1.3 Routing / 13	
1.3.1 Routing across Providers / 14	
1.3.2 Routing within Edge Networks / 15	
1.3.3 Routing Scalability / 16	
References / 18	
2 Transport-Layer Protocols	19
2.1 Transmission Control Protocol / 19	
2.1.1 Transmission Control Protocol Header Structure / 19	
2.1.2 Three-Way Handshake / 20	
2.1.3 Transmission Control Protocol Flow Control and Congestion Control / 21	
2.1.4 Port Number / 24	

2.2	User Datagram Protocol / 25
2.2.1	User Datagram Protocol Header Structure / 25
2.3	Stream Control Transmission Protocol / 26
2.3.1	Stream Control Transmission Protocol Packet Structure / 26
2.3.2	Security: Prevention of SYN Attacks / 27
2.4	Real-Time Transport Protocol / 29
2.4.1	Real-Time Transport Protocol Header Structure / 29
	References / 30

3 Internet Architecture 31

3.1	Internet Exchange Point / 31
3.2	History of Internet Exchange Points / 33
3.3	Internet Service Provider Interconnection Relationships / 34
3.4	Peering and Transit / 35
	References / 37

4 IP Routing Protocols 39

4.1	Overview of Routing Protocols / 40
4.1.1	Interior Gateway Protocol / 41
4.1.2	Exterior Gateway Protocol / 42
4.2	Routing Information Protocol / 43
4.2.1	Routing Information Protocol Header Format / 43
4.2.2	Update of Routing Table in Routing Information Protocol / 44
4.2.3	Maintenance of Routing Table in Routing Information Protocol / 46
4.2.4	Split Horizon / 47
4.2.5	Limitations of Routing Information Protocol / 47
4.3	Open Shortest Path First / 48
4.3.1	Shortest-Path Algorithm / 48
4.3.2	Hierarchical Routing / 51
4.3.3	Open Shortest Path First Packet Format / 51
4.3.4	Comparison of Routing Information Protocol and Open Shortest Path First / 52
4.4	Border Gateway Protocol / 53
4.4.1	Border Gateway Protocol Message Flows / 53
4.4.2	Border Gateway Protocol Policy Selection Attributes / 54
	References / 57

7.3.4	Protocol-Independent Multicast / 101	
7.3.5	Simple Multicast Routing Protocol / 101	
7.4	Anycasting / 102	
7.4.1	Architectural Issues / 103	
7.4.2	Anycast Addresses / 103	
7.4.3	Differences between the Services Offered by IP Multicasting and IP Anycasting / 104	
7.5	IPv6 Anycast Routing Protocol: Protocol-Independent Anycast—Sparse Mode / 105	
	References / 106	
8	Layer-2 Transport over Packet	109
8.1	Draft-Martini Signaling and Encapsulation / 109	
8.1.1	Functionality / 110	
8.1.2	Encapsulation / 110	
8.1.3	Protocol-Specific Encapsulation / 111	
8.2	Layer-2 Tunneling Protocol / 114	
8.2.1	Layer-2 Tunneling Protocol Version 3 / 115	
8.2.2	Pseudowire Emulation Edge to Edge / 118	
	References / 121	
9	Virtual Private Wired Service	123
9.1	Types of Private Wire Services / 123	
9.1.1	Layer-2 Virtual Private Services: Wide Area Networks and Local Area Networks / 124	
9.1.2	Virtual Private Wire Service / 126	
9.1.3	Virtual Private Multicast Service / 127	
9.1.4	IP-Only Layer-2 Virtual Private Network / 128	
9.1.5	Internet Protocol Security / 129	
9.2	Generic Routing Encapsulation / 130	
9.3	Layer-2 Tunneling Protocol / 131	
9.4	Layer-3 Virtual Private Network 2547bis, Virtual Router / 131	
9.4.1	Virtual Router Redundancy Protocol / 133	
	References / 136	
10	IP and Optical Networking	137
10.1	IP/Optical Network Evolution / 138	
10.1.1	Where Networking Is Today / 138	
10.1.2	Where Networking Is Going / 139	

10.2	Challenges in Legacy Traditional IP/Optical Networks / 140	
10.2.1	Proprietary Network Management Systems / 140	
10.2.2	Complexity of Provisioning in Legacy IP/Optical Networks / 141	
10.3	Automated Provisioning in IP/Optical Networks / 142	
10.4	Control Plane Models for IP/Optical Networking / 144	
10.4.1	Optical Internetworking Forum's Optical User Network Interface: Overlay Model / 145	
10.4.2	Internet Engineering Task Force's Generalized Multiprotocol Label Switching: Peer Model / 145	
10.5	Next-Generation MultiLayer Network Design Requirements / 147	
10.6	Benefits and Challenges in IP/Optical Networking / 148	
	References / 149	
11	IP Version 6	151
11.1	Addresses in IP Version 6 / 152	
11.1.1	Unicast IP Addresses / 152	
11.1.2	Multicast IP Addresses / 153	
11.2	IP Packet Headers / 154	
11.3	IP Address Resolution / 155	
11.4	IP Version 6 Deployment: Drivers and Impediments / 156	
11.4.1	Need for Backwards Compatibility / 157	
11.4.2	Initial Deployment Drivers / 158	
11.4.3	Reaching a Critical Mass / 160	
	References / 161	
12	IP Traffic Engineering	163
12.1	Models of Traffic Demands / 163	
12.2	Optimal Routing with Multiprotocol Label Switching / 165	
12.2.1	Overview / 165	
12.2.2	Applicability of Optimal Routing / 165	
12.2.3	Network Model / 166	
12.2.4	Optimal Routing Formulations with Three Models / 166	
12.3	Link-Weight Optimization with Open Shortest Path First / 169	
12.3.1	Overview / 169	
12.3.2	Examples of Routing Control with Link Weights / 170	
12.3.3	Link-Weight Setting Against Network Failure / 172	
12.4	Extended Shortest-Path-Based Routing Schemes / 173	
12.4.1	Smart-Open Shortest Path First / 174	

- 12.4.2 Two-Phase Routing / 174
- 12.4.3 Fine Two-Phase Routing / 176
- 12.4.4 Features of Routing Schemes / 177
- References / 177

13 IP Network Security 181

- 13.1 Introduction / 181
- 13.2 Detection of Denial-of-Service Attack / 182
 - 13.2.1 Backscatter Analysis / 182
 - 13.2.2 Multilevel Tree or Online Packet Statistics / 184
- 13.3 IP Traceback / 187
 - 13.3.1 IP Traceback Solutions / 189
- 13.4 Edge Sampling Scheme / 189
- 13.5 Advanced Marking Scheme / 193
- References / 196

14 Mobility Support for IP 197

- 14.1 Mobility Management Approaches / 199
 - 14.1.1 Host Routes / 200
 - 14.1.2 Tunneling / 201
 - 14.1.3 Route Optimization / 203
- 14.2 Security Threats Related to IP Mobility / 205
 - 14.2.1 Impersonation / 205
 - 14.2.2 Redirection-Based Flooding / 208
 - 14.2.3 Possible Solutions / 210
- 14.3 Mobility Support in IPv6 / 213
- 14.4 Reactive Versus Proactive Mobility Support / 218
- 14.5 Relation to Multihoming / 219
- 14.6 Protocols Supplementing Mobility / 220
 - 14.6.1 Router and Subnet Prefix Discovery / 220
 - 14.6.2 Movement Detection / 221
 - 14.6.3 IP Address Configuration / 222
 - 14.6.4 Neighbor Unreachability Detection / 223
 - 14.6.5 Internet Control Message Protocol for IP Version 6 / 224
 - 14.6.6 Optimizations / 224
 - 14.6.7 Media-Independent Handover Services / 227
- References / 231

Index 235

PREFACE

Today, the Internet and computer networking are essential to business, learning, personal communication, and entertainment. The infrastructure that carries virtually all messages and transactions sent over the Internet is based on advanced Internet protocols. These advanced Internet protocols ensure that both public and private networks operate with maximum performance, security, and flexibility.

This book is intended to provide a comprehensive technical overview and survey of advanced Internet protocols. First, a solid introduction and discussion of internet-working technologies, architectures, and protocols is provided. The book also presents the application of these concepts into next-generation networks and discusses protection and restoration as well as various tunneling protocols and applications. Finally, emerging topics are discussed.

This book covers basic concepts in Transmission Control Protocol (TCP)/Internet Protocol (IP), Internet architecture, IP routing protocols including transport-layer protocols, IP version 6, Multiprotocol Label Switching (MPLS), networking services such as IP Quality of Service (QoS), IP Multicast, anycast, Layer-2 and Layer-3 virtual private networks (L2VPN and L3VPN), and applications such as IP over Dense Wavelength Division Multiplexing (DWDM), IP traffic engineering, IP in mobility, and IP network security.

Although there are no strict prerequisites for reading this book, a data communication background would be helpful. All the concepts in this book are developed from basics on an intuitive basis, with further insight provided through examples of real-world networks, services, and applications. The authors are well-experienced researchers and engineers from both industry and academia.

Dr. Mallikarjun Tatipamula first offered a course on Advanced Internet Protocols to graduate students at a number of leading universities in addition to tutorials at various conferences. The positive response from students triggered the idea of writing this book with the objective of setting a strong foundation for students regarding Internet protocols.

The authors have developed some of the contents of this book in their graduate courses and seminars in their universities and organizations. These contents have been improved, thanks to feedback from students and industry colleagues. The courses in which this material has been used have attracted both academic and industrial practitioners, as the Internet and computer networking are key topics in the information

technology industry. These are people interested in the principles of the Internet and advanced networking technologies, including designing networks and network elements and being able to consult network designers in order to satisfy their customers' needs.

Audience

This book is intended for graduate students, R&D managers, software and hardware engineers, system engineers, and telecommunications and networking professionals. This book will interest those who are currently active or anticipate future involvement in internetworking and are seeking a broad understanding and comprehensive technical overview of Internet technologies, architectures, and protocols including current status and future direction.

Organization

The book is organized as follows.

- Chapter 1 describes Transmission Control Protocol (TCP)/Internet Protocol (IP), which is an Internet protocol stack that enables communications between two computers, or hosts, through the Internet. It is a collection of different protocols. A protocol is a set of rules that controls the way data is sent between hosts.
- Chapter 2 explains protocols in the transport layer, which is the fourth layer of the Open Systems Interconnection (OSI) reference model. Transparent transfer of data between end users using services of the network layer is provided. The well-known protocols in this layer are TCP, User Datagram Protocol (UDP), Stream Control Transmission Protocol (SCTP), and Real-time Transport Protocol (RTP).
- Chapter 3 describes Internet architecture, including basic Internet topology, Internet exchange points (IXPs), the history of IXPs, and the principles of Internet relationships and Internet service providers (ISPs).
- Chapter 4 describes IP routing protocol including an overview of Interior Gateway Protocols (IGPs) and Exterior Gateway Protocols (EGPs). Routing Information Protocol (RIP) and Open Shortest Path First (OSPF) is covered as an IGP, and Border Gateway Protocol (BGP) is covered as an EGP.
- Chapter 5 describes Multiprotocol Label Switching (MPLS) technologies, which enable networks to perform traffic engineering, resident communications, virtual private networks, Ethernet emulation, and so on. First, an overview to MPLS is given. Next, the functions and mechanisms of MPLS are described. Finally, MPLS applicabilities are discussed.

- Chapter 6 presents a myriad of concepts and paradigms used to provision quality of service by the Internet. It includes discussions on the implementation of mechanisms of traffic differentiators and policy and traffic policers and regulators. The discussion includes Diffserv, IntServ, and a combination of both as developed by recent research.
- Chapter 7 describes IP multicast and anycast. The majority of the traffic on the Internet is *unicast*: one source device sending to one destination device. However, where the number of receivers is more than one, multicast traffic can be transmitted over the Internet in ways that avoid loading the network with excessive traffic. The multicast mechanisms present the possibility of having multiple receivers for routing and packet delivery.
- Chapter 8 presents several Layer-2 encapsulation protocols. These encapsulation protocols are used to allow Layer-2 connectivity through nonadjacent networks. These protocols allow remote networks to communicate as if they were in the same network. Several examples of these encapsulation protocols are presented including Martini, Layer-2 Transport, and Pseudowire Emulation Edge to Edge protocol.
- Chapter 9 introduces point-to-point virtual connectivity and virtual broadcast access connectivity, such as virtual local area networks (LANs). Layer-2 virtual private network over transport can provide several features such as redundancy against failures and controlled connectivity through a packet-switched network or Generalized MPLS (GMPLS). Multiple Layer-2 access protocols can be virtually emulated.
- Chapter 10 introduces the multilayer network evolution, followed by discussion on limitations in current legacy networks, automated provisioning in IP/optical networks, the proposed control plane techniques in the industry, key design requirements for next-generation multilayer IP/optical networks, and comparisons of the proposed control plane models against these key design requirements.
- Chapter 11 describes the main differences between IPv4 and IPv6, investigates the reasons for the lagging uptake of IPv6 so far, and explains why, contrary to widespread belief, IPv6's large address space alone is insufficient to drive early IPv6 deployment.
- Chapter 12 describes several routing schemes to maximize the network utilization for various traffic demand models for MPLS and OSPF technologies. One useful approach to enhancing routing performance is to minimize the maximum link utilization rate, also called the network congestion ratio, of all network links. Minimizing the network congestion ratio leads to an increase in admissible traffic.
- Chapter 13 discusses some popular threats and how they exploit protocol vulnerabilities. Some counter measures, based on additional algorithms that either work as an application or as a protocol at the network or transport layers, are discussed including methods to trace back threatening hosts.

- Chapter 14 explains the different ways of augmenting the Internet for mobility support, as well as the security threats that need to be considered as part of this. The relationship between mobility and multihoming will be explored further. The chapter also gives an overview of auxiliary protocols that play a fundamental role in IP mobility management.

ELJI OKI
ROBERTO ROJAS-CESSA
MALLIKARJUN TATIPAMULA
CHRISTIAN VOGT

ACKNOWLEDGMENTS

This book could not have been published without the help of many people. We thank them for their efforts in improving the quality of the book. We have done our best to accurately describe advanced Internet protocols, services, and applications as well as the basic concepts. We alone are responsible for any remaining errors. If any error is found, please send an e-mail to eiiji.oki@uec.ac.jp and rojas@njit.edu. We will correct them in future editions.

Several chapters of the book are based on our research works. We would like to thank the people who have contributed materials to some chapters, especially Dr. Nattapong Kitsuwon (UEC Tokyo), Mohammad Kamrul Islam (UEC Tokyo), Khondaker M. Salehin (New Jersey Institute of Technology), and Tuhina Sarkar (New Jersey Institute of Technology).

The entire manuscript draft was reviewed by Katsuhiro Amako (UEC Tokyo), Dr. Neda Beheshti (Ericsson), Komlan Egoh (New Jersey Institute of Technology), Prof. Ziqian Dong (New York Institute of Technology), Agostinho A. Jose (UEC Tokyo), Prof. Chuan-Bi Lin (Chaoyang University of Technology, Taiwan), Abu Hena Al Muktedir (UEC Tokyo), Ihsen Aziz Ouédraogo (UEC Tokyo), Dr. Kiran Yedavalli (Ericsson), and Dr. Ying Zhang (Ericsson). We are immensely grateful for their comments and suggestions.

Eiji wishes to thank his wife, Naoko, his daughter, Kanako, and his son, Shunji, for their love and support. Roberto would like to thank his wife, Vatcharapan, and his children, Marco and Nelli, for their unconditional understanding, love, and support. Mallikarjun is grateful to his wife, Latha, and his children, Sashank, Santosh, and Vaishnavi, for their love and support.

EIJI OKI
ROBERTO ROJAS-CESSA
MALLIKARJUN TATIPAMULA
CHRISTIAN VOGT

ABOUT THE AUTHORS

Eiji Oki is an Associate Professor at the University of Electro-Communications, Tokyo, Japan. He received his Bachelor and Master of Engineering degrees in Instrumentation Engineering and a Doctorate of Philosophy in Electrical Engineering from Keio University, Yokohama, Japan, in 1991, 1993, and 1999, respectively. In 1993, he joined Nippon Telegraph and Telephone Corporation (NTT) Communication Switching Laboratories, Tokyo, Japan. He has been researching network design and control, traffic-control methods, and high-speed switching systems. From 2000 to 2001, he was a Visiting Scholar at the Polytechnic Institute of New York University, Brooklyn, New York, where he was involved in designing terabit switch/router systems. He was engaged in researching and developing high-speed optical IP backbone networks with NTT Laboratories. He joined the University of Electro-Communications, Tokyo, Japan, in July 2008. He has been active in standardization of path computation element (PCE) and GMPLS in the Internet Engineering Task Force (IETF). He wrote 11 IETF RFCs. He served as a guest co-editor for the special issue on “Multi-Domain Optical Networks: Issues and Challenges,” June 2008, in IEEE Communications Magazine; a guest co-editor for the special issue on Routing, “Path Computation and Traffic Engineering in Future Internet,” December 2007, in the Journal of Communications and Networks; a guest co-editor for the special section on “Photonic Network Technologies in Terabit Network Era,” April 2011, in IEICE Transactions on Communications; a Technical Program Committee (TPC) Co-Chair for the Workshop on High-Performance Switching and Routing in 2006 and 2010; a Track Co-Chair on Optical Networking for ICCCN 2009; a TPC Co-Chair for the International Conference on IP+Optical Network (iPOP 2010); and a Co-Chair of Optical Networks and Systems Symposium for IEEE ICC 2011. Professor Oki was the recipient of the 1998 Switching System Research Award and the 1999 Excellent Paper Award presented by IEICE, the 2001 Asia-Pacific Outstanding Young Researcher Award presented by IEEE Communications Society for his contribution to broadband network, ATM, and optical IP technologies, and the 2010 Telecom System Technology Prize by the Telecommunications Advanced Foundation. He has co-authored two books, *Broadband Packet Switching Technologies*, published by John Wiley & Sons, Inc., New York, in 2001, and *GMPLS Technologies*, published by CRC Press, Boca Raton, Florida, in 2005. He is an IEEE Senior Member.

Roberto Rojas-Cessa received a Master of Computer Engineering degree and a Doctorate of Philosophy in Electrical Engineering from the Polytechnic Institute of New York University, Brooklyn, New York. He also received a Master of Science in Electrical Engineering from the Research and Advanced Studies Center (CIVESTAV) in Mexico. He received his Bachelor of Science in Electronic Instrumentation from Universidad Veracruzana, Mexico. Currently, he is an Associate Professor in the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, Newark, New Jersey. He was an Adjunct Professor and a Research Associate in the Department of Electrical and Computer Engineering of Polytechnic Institute of New York University. He has been involved in design and implementation of application-specific integrated-circuits (ASIC) for biomedical applications and high-speed computer communications, and in the development of high-performance and scalable packet switches and reliable switches. He was part of the team designing a 40 Tb/s core router at Corec, Inc. in Tinton Falls, NJ. His research interests include high-speed switching and routing, fault tolerance, quality-of-service networks, network measurements, and distributed systems. His research has been funded by the U.S. National Science Foundation and Industry. He was the recipient of the Advance in Research Excellence of the ECE Department in 2004. He has served on several technical committees for IEEE conferences and as a reviewer for several IEEE journals. He has been a reviewer and panelist for the U.S. National Science Foundation and the U.S. Department of Energy. He has more than 10 years of experience in teaching Internet protocols and computer communications. Currently, he is the Director of the Networking Research Laboratory at the ECE Department and the Coordinator of the Networking Research Focus Area Group of the same department.

Mallikarjun Tatipamula is Head of Packet Technologies Research, Ericsson Silicon Valley. He leads a research team and is responsible for innovation and implementation of leading-edge technologies including Openflow, next-generation routing architectures, application-aware networking, and Cloud computing, networking, and services. He closely works with leaders from universities, research and education networks, and service providers around the world. Prior to his role at Ericsson, he was Vice President and Head of Service Provider Sector at Juniper Networks, Sunnyvale, California. His team responsibilities include new technologies, architectures, standards, solutions, and creation of business strategy for the implementation of next-generation products in content delivery network/video, Cloud, IP/optical integration, security, mobility, and convergence. Prior to Juniper, he was with Cisco systems for over eight years and made active contributions to the Cisco IP NGN strategy. These contributions include “Service Exchange Framework”, IMS/FMC, advanced technologies such as IPv6, GMPLS, multicast, along with his contributions in the early days of VoIP, mobile wireless and IP/optical integration that led to definition and implementation plans. Prior to Cisco, Mallikarjun was a principal engineer at Motorola in the Cellular Infrastructure Group. He was responsible for defining system architecture for advanced wireless and satellite systems. From 1993 to 1997 he was a senior member of the scientific staff at BNR

(now Nortel), Ottawa, responsible for development of Nortel's optical (OC12/OC48) products and involved in the development of the Nortel CDMA Base Station. From 1990 to 1992, Mallikarjun was with Indian Telephone Industries as an Assistant Executive Engineer in Optical Transmission R&D Laboratories and Indian Institute of Technology, Chennai, as a Senior Project Officer in the Fiber Optics Labs, responsible for development of optical data links. He is a key note speaker at leading telecommunications and networking events. Mallikarjun obtained Doctorate of Philosophy in Information Science and Technology from the University of Tokyo, Japan, a Master of Science in Communication Systems and High Frequency Technologies from the Indian Institute of Technology, Chennai, and a Bachelor of Technology in Electronics and Communication Engineering from NIT, Warangal, India. Mallikarjun has authored or coauthored a number of patents and publications with leaders from NRENs and service providers. He is a trusted partner, advisor, and thought leader. Mallikarjun is a lead editor for "Multimedia Communication Networks: Technologies and Services," by Artech House Publishers. He is a senior member of IEEE and his biography appeared in Marquis Who is Who in the World, Who is Who in America, and Who is Who Science and Engineering. Mallikarjun delivered distinguished lectures at leading universities including Stanford University, the University of Tokyo, the Tokyo Institute of Technology, Tsing Hua University, Beijing University, China, and IIT Delhi.

Christian Vogt is a Senior Marketing Manager at Ericsson Silicon Valley. He works on product and marketing strategies regarding the convergence of wireless and wireline networks, and their transition from IP version 4 to 6. His technical interests further include Internet routing and addressing, IP mobility and multihoming, related security aspects, as well as next-generation Internet architectures. As part of this work, Christian co-chairs the Internet Area and Source Address Validation Improvements working groups in the Internet Engineering Task Force. Christian received his doctoral degree from the University of Karlsruhe in Germany in 2007 for his dissertation on efficient and secure mobility support in IP version 6. He also holds a Master of Science in Computer Science from the University of Southern California, Los Angeles, and a German Diplom in Computer Science from the University of Bonn. Currently, Christian is pursuing an Executive Master of Business Administration degree at the Wharton School, University of Pennsylvania.

TRANSMISSION CONTROL PROTOCOL/INTERNET PROTOCOL OVERVIEW

This first chapter provides an overview of Transmission Control Protocol (TCP)/Internet Protocol (IP), which is an Internet protocol stack to perform communications between two computers, or hosts, through the Internet. It is a collection of different protocols. A *protocol* is a set of rules that controls the way data is transmitted between hosts.

1.1 FUNDAMENTAL ARCHITECTURE

Figure 1.1 shows the Open Systems Interconnection (OSI) reference model and the Internet protocol stack. The International Standardization Organization (ISO) specifies a guideline called the OSI reference model. It is an abstract description for layered communications and computer network protocol design. It consists of seven layers, which are, from the bottom, physical, data link, network, transport, session, presentation, and application, as shown in Figure 1.1a. The application layer is the OSI layer closest to the end user, which means that both the OSI application layer and the user interact directly with the software application. This layer interacts with software applications that implement a communicating component. At the physical layer, data is recognized, or handled, as bits. At the data link layer, data is handled as frames. At the network layer, data is recognized as packets. In the transport layer, data is handled as segments or datagrams. In the remaining upper layers, users' information is recognized.

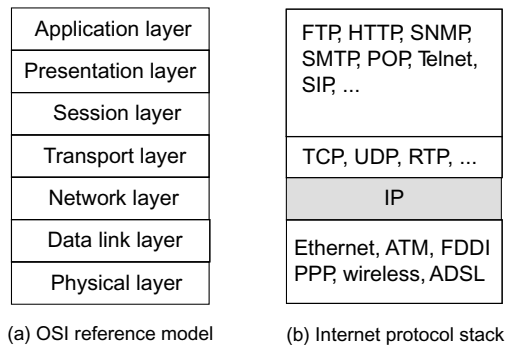


Figure 1.1 Layer model.

The Internet protocol stack is shown in Figure 1.1b, where we can see to which layer in the OSI reference model each protocol corresponds. For example, the Internet protocol corresponds to the network layer. TCP corresponds to the transport layer. The Internet protocol stack that includes several protocols, as shown in Figure 1.1b, is referred to as *TCP/IP*.

TCP/IP is being standardized by the Internet Engineering Task Force (IETF). It is widely used as the *de facto* standard protocol for building network equipment and internetworking. If the TCP/IP standard is used, computers can communicate with each other regardless of hardware and operating system.

Figure 1.2 shows a basic philosophy of TCP/IP. In the TCP/IP philosophy, the IP core network functions simply and quickly, while the functionality of the edges surrounding the core is complex. Edges can be hosts, edge routers, network boundaries, etc. The IP network based on this philosophy is scalable and flexible. This enables different complexities at the edge.

Application data and customer traffic can be transmitted on the IP layer because IP is the least common denominator, as shown in Figure 1.3. IP supports transport-layer protocols such as TCP, User Datagram Protocol (UDP), Real Time Protocol (RTP)/UDP, and Stream Control Transmission Protocol (SCTP), as shown in

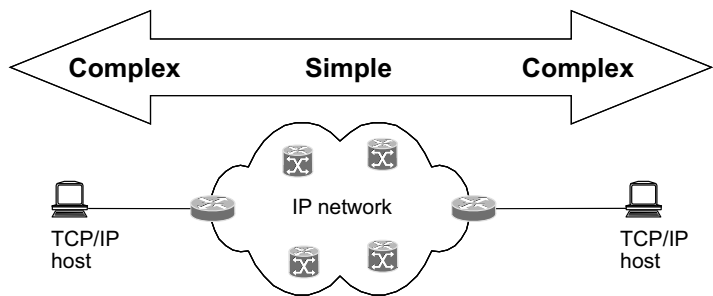


Figure 1.2 Basic philosophy of TCP/IP.

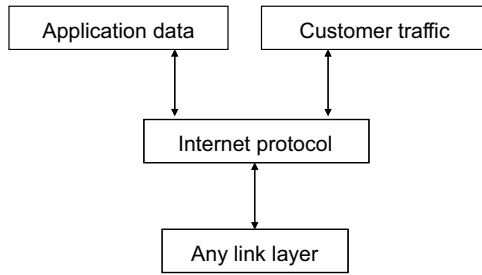


Figure 1.3 IP is the least common denominator.

Figure 1.4. IP works on link-layer protocols such as Ethernet, Point-to-Point Protocol (PPP) over Asynchronous Transer Mode (ATM), satellite, wireless, optical, and IP/Multiprotocol Label Switching (MPLS), as shown in Figure 1.5.

More than one network can be connected via IP, as shown in Figure 1.6. IP enables building a large-scale network where smaller networks, or subnetworks, are concatenated to build one large network. Inside each network, a node does not need to be connected with other nodes in the same network via IP. This is the way the Internet is built.

Figure 1.7 shows the protocol stack based on the OSI reference model and network equipment and protocols connecting layers. A repeater is a functional device of layer 1. It enhances signal level. A bridge, or switch, processes frames, for example Ethernet frames, corresponding to layer 2. Two bridges are connected via Ethernet protocol. A

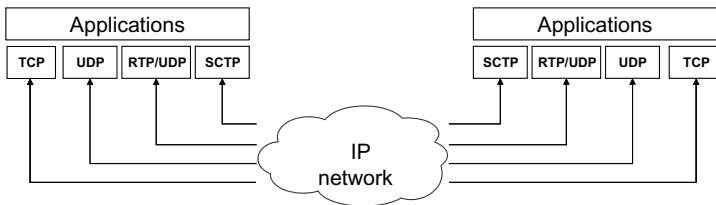


Figure 1.4 Any transport-layer protocols on IP.

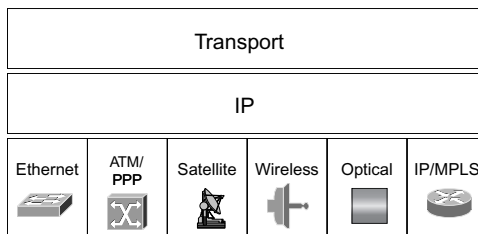


Figure 1.5 Any link-layer protocols under IP.

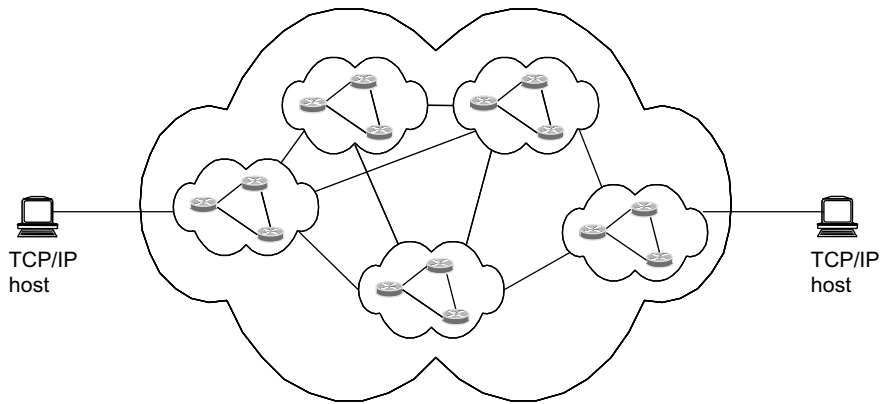


Figure 1.6 Interworking via IP.

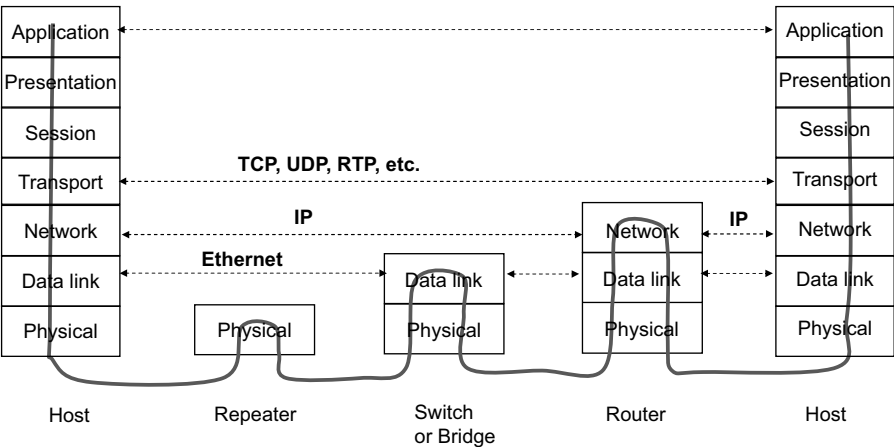


Figure 1.7 Protocol stack and connectivity.

router processes IP packets corresponding to layer 3. Two routers are connected via IP. Two hosts are connected via protocols for layers 4, 5, 6, and 7. For example, TCP, UDP, and RPT are used to connect two hosts on layer 4.

1.2 INTERNET PROTOCOL BASICS

Internet Protocol is a protocol in the network layer. It is used for sending data from a source host to its destination host where each host has a unique number. The IP header format is shown in Figure 1.8. Version specifies the IP version. Header length indicates the size of the header. Type of service is used to guide the selection of the